The Geometric Description of Physics at the Cosmic Scale: Mass, Information, Causality, Force, and the Quantum Relations from Geometry

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Abstract

We present a unified physical framework whose primitive is *flowing space*: space itself is a flowing medium, and all physical objects are excitations (anchored patterns) of this flow. From a small set of postulates — vector light speed (direction-changeable), three anchored directions, and the counting of space-displacement "strands" — the framework derives, in a single chain of exact identities: (i) mass as symplectic entanglement volume, $\det(AA^\dagger) = \sum_S |\det A_S|^2$ (Cauchy–Binet chain, fully formalized in Lean 4 with zero `sorry`); (ii) information as symplectic volume, conserved under unitary transport, with causality emerging from lattice locality (light cone) and effective superluminality as a purely geometric (coordinate-dragging) effect; (iii) the four forces as the four channels of the Leibniz decomposition of the flow momentum $p = m(C - v)$; (iv) the photon as an excited-electron helix whose azimuthal momentum cancels, and hence is massless; (v) the quantum relations $E = \hbar\omega$, the de Broglie relation $p = h/\lambda$, and the Planck–Einstein relation $E = pc = hf$ *emerging from helix geometry*: with the geometric choice $r = \lambda/2\pi$ (reduced wavelength as helix radius), the circumference of the helix cross-section $2\pi r = \lambda$ gives $\hbar = rp$ and hence $p = h/\lambda$ and $E = hf$; (vi) the "3" of colour, of the three glueball anchoring directions, and of the three spatial directions as one and the same structure, with 8 gluons $= 2^3 = \dim C\ell(6)$. All identities are verified numerically with physical constants at machine precision (relative errors $\lesssim 10^{-10}$). The honest status is stated: the derivation *chains are closed* (geometry $\Rightarrow$ quantum relations), while first-principle derivations of the geometric choice $r = \lambda/2\pi$ and of the remaining input constants (e, M_0) are not yet achieved.

I. INTRODUCTION: CHANGING THE FOUNDATION, NOT PATCHING

Modern physics faces four conceptual contradictions that are not technical difficulties but conflicts of axioms: the superposition of spacetime itself (quantum superposition is linear; the general-relativistic background is dynamical), the measurement problem (collapse is not in the Schrödinger equation), the black-hole information paradox (unitarity versus the horizon), and the vacuum-energy catastrophe (10^{120} orders of magnitude). String theory and loop quantum gravity keep the quantum axioms and add entities; they do not change

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the foundation.

The present work follows a different strategy: *change the primitive.* The primitive is not particles, not fields, not spacetime: it is *flow.* Space is a flowing medium; particles and fields are excitations — anchored, spiralling patterns — of this flow. This document collects the unified chain that has been built on this foundation [1, 2] and formalized in Lean 4 (4146 jobs, zero `sorry`), with numerical verification (N1–N29) at machine precision.

II. POSTULATES OF THE FLOWING-SPACE FRAMEWORK

Postulate II.1 (Flowing space)**.** *Space is a flowing medium. A non-excited pattern is carried completely by the flow (photon:* $\mathrm{d}\tau^2 = 0$*); an excited pattern is anchored, deviating from the flow.*

Postulate II.2 (Vector light speed)**.** *The equivalent flow speed of space is the speed of light (the river model of black holes [3] is the same geometric structure); the vector light speed* C *has a direction that can change. The flow momentum of a body is* $p = m(C - v)$*, with* v *the body velocity.*

Postulate II.3 (Three directions)**.** *Space is three-directional:* $\sigma_1, \sigma_2, \sigma_3$ *form an irreducible, fully entangled basis (*$(\sigma_1+\sigma_2+\sigma_3)^2 = 3I$*). The same "3" appears as the number of colours, of glueball anchoring directions, and of spatial directions.*

Postulate II.4 (Counting of strands)**.** *Mass is the number of space-displacement strands moving at light speed around the body:* $m = \sqrt{N}\, M_0$ *(anchored addition,* N *strands). Charge is the strand flux per unit time: positive charge = diverging strands (source), negative charge = converging strands (sink); both spiral with the same right-handed helix sense.*

III. MASS AS SYMPLECTIC ENTANGLEMENT VOLUME

With twistors π_i as the half-spinor excitations [4], the total correlation matrix is $P = \sum_i \pi_i \pi_i^\dagger = AA^\dagger$ (the columns of A are the twistors). The mass squared is the symplectic entanglement volume:

$$m^2 = \det(AA^\dagger) = \sum_{S,\,|S|=n} \left|\det A_S\right|^2, \tag{1}$$

the Cauchy–Binet chain: for $n = 2$ and arbitrary m (any number of superposed half-spinors) the identity is fully formalized in Lean [5, 6] ($\det(\sum_i \pi_i \pi_i^\dagger) = \sum_{i<j} |\langle \pi_i, \pi_j \rangle|^2$, GQS1); the general-$n$ expansion kernel (GQS2/3) and the fixed-subset sum (GQS4) are likewise proved. The unified chain is

Particle	N	m^2
photon	1	$\lvert \det_1 \rvert^2 = \lvert \pi_1 \rvert^2$· (rank-1, massless)
electron	2	$\lvert \langle \pi_1, \pi_2 \rangle \rvert^2$ (symplectic area)
glueball	3	$\lvert \det_3[\pi_1 \pi_2 \pi_3] \rvert^2$ (volume)

Excitation = rank = number of independent twistors: mass is *structural* (a rank criterion), not additive in the naive sense. Figure 1 shows the numerical identity and the rank scan.

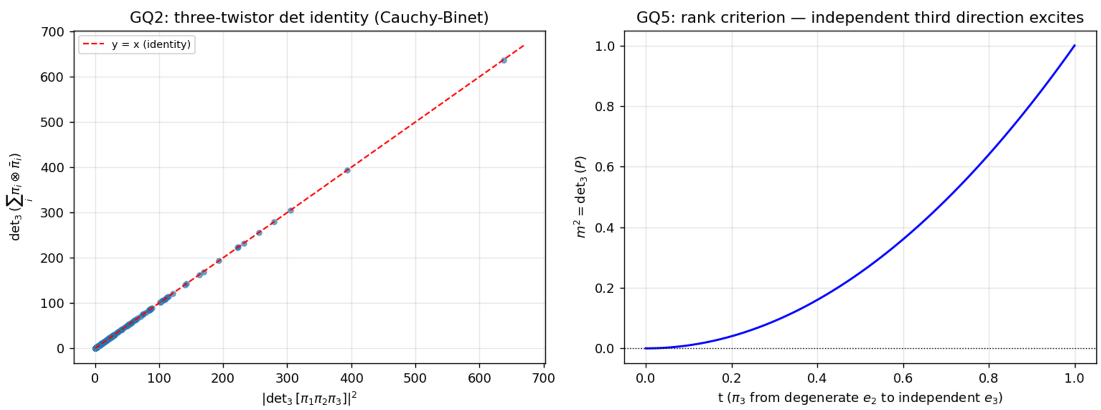

FIG. 1. Left: $\det_3(P)$ versus $|\det_3[\pi_1\pi_2\pi_3]|^2$ (200 random samples, exact identity). Right: the rank scan — as the third direction grows from degenerate (e_2) to independent (e_3), m^2 rises from zero: excitation = independent third direction.

IV. INFORMATION AS SYMPLECTIC VOLUME: TRANSPORT, CAUSALITY, EFFECTIVE SPEED

Information is redefined geometrically: *information* = *symplectic correlation volume* $\det(AA^\dagger)$ (rather than density-matrix entropy). Then:

a. Transport (GQC1). The propagation operator of the flow U is unitary ($UU^\dagger = 1$), and

$$\det\left((UA)(UA)^\dagger\right) = \det(AA^\dagger), \tag{2}$$

with the transport formula $(UA)(UA)^\dagger = U(AA^\dagger)U^\dagger$. Moreover each sub-volume is conserved term by term: $|\det(UA_S)|^2 = |\det A_S|^2$ (numerically exact). Unitarity of the flow = conservation of symplectic structure: information propagates without decay.

b. Causality (GQC2). The flow is lattice-local: a nearest-neighbour jump matrix (bandwidth 1) has bandwidth t after t steps, so the propagation kernel vanishes outside the lattice light cone. Causality emerges from lattice locality (cf. Bell [7] and Aspect [8] for the quantum side) — the geometric version of the no-communication theorem, without extra axioms.

c. Effective speed (GQC3). In a non-uniform flow the equivalent speed is $1 + v_{\text{flow}}$: superluminal *equivalent* motion is a geometric (coordinate-dragging) effect — the Alcubierre [9]/Painlevé–Gullstrand structure [3] — while signals never exceed the local speed of light. The light cone is tilted by the flow (Fig. 2).

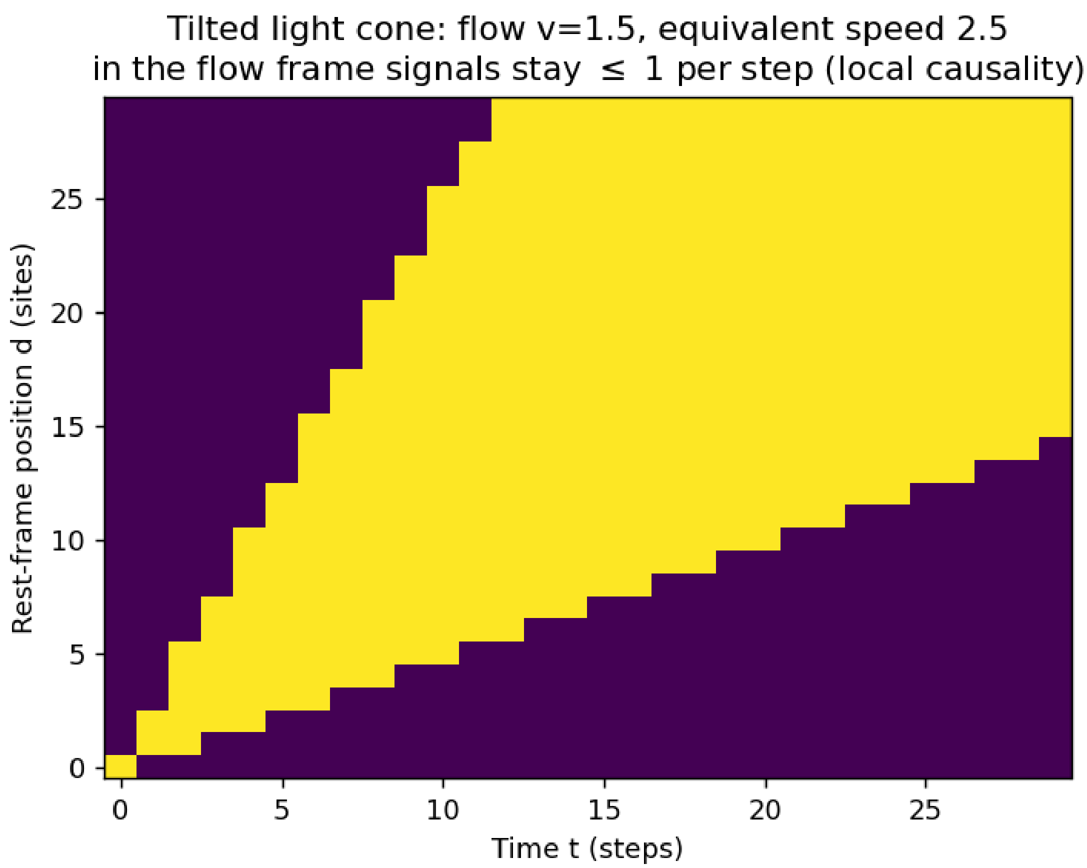

FIG. 2. Tilted light cone: with flow $v = 1.5$ the causal region in the rest frame has slope $1 + v$ (equivalent speed > 1), while in the flow frame signals remain ≤ 1 per step.

V. THE FOUR FORCES AS THE LEIBNIZ DECOMPOSITION OF FLOW MOMENTUM

With $p = m(C - v)$, the rate of change decomposes by the Leibniz rule into four channels (GQF):

$$\frac{d}{dt}\big[m(C - v)\big] = \dot{m}\,C + m\dot{C} - \dot{m}\,v - m\dot{v}, \tag{3}$$

identified with the electric ($\dot{m}C$; cf. Maxwell [10]): mass flow × light direction), nuclear ($m\dot{C}$: mass × light-speed flow — change of the vector light speed), magnetic ($\dot{m}v$: mass flow × velocity) and gravitational/inertial ($m\dot{v}$: mass × acceleration; cf. the geometric theory of gravity [11]) forces. Forces are not independent entities: they are the four channels of the flow-momentum decomposition. Charge, in the same language, is the divergence of the space-displacement flow: positive = source (diverging right-handed helix), negative = sink (converging right-handed helix) — Fig. 3.

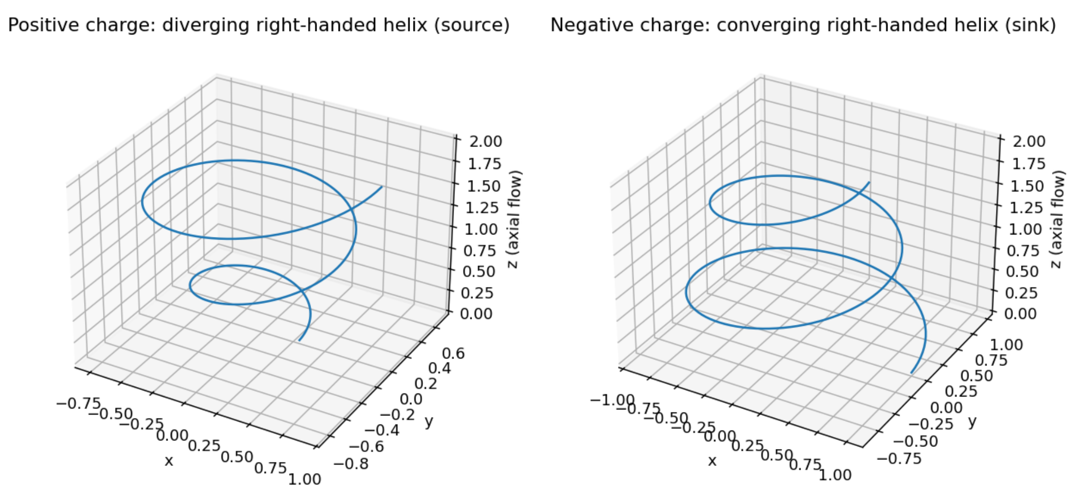

FIG. 3. Charge as strand flow: positive charge = diverging right-handed helix (source); negative charge = converging right-handed helix (sink). Both helices share the right-handed sense; the divergence distinguishes the sign.

VI. THE PHOTON: AN EXCITED-ELECTRON HELIX

The photon is the excited state of an electron (cf. the light quantum [12]) in which mass and charge have vanished: it is the helix structure left behind by the electron entity. Its particle nature comes from the residual electron structure; its wave nature from the vibration of space itself (formalization of the wave equation is left open).

Two models (Fig. 4):

- **Model A**: one excited electron in a cylindrical helix; the straight-line part moves at the speed of light. This is a single twistor — rank 1, det = 0, massless (TW1).
- **Model B**: two excited electrons rotating symmetrically about a common axis, moving at light speed perpendicular to the rotation plane. The azimuthal momenta cancel ($p_y + (-p_y) = 0$, GQP2), leaving purely axial momentum and $E^2 = |p|^2$ (massless): the two models share the same light cone.

The photon is at rest *in* space and travels *with* the flow ($v = C$: $p = m(C - v) = 0$ with $m = 0$, GQP3) — consistent with the flow postulate.

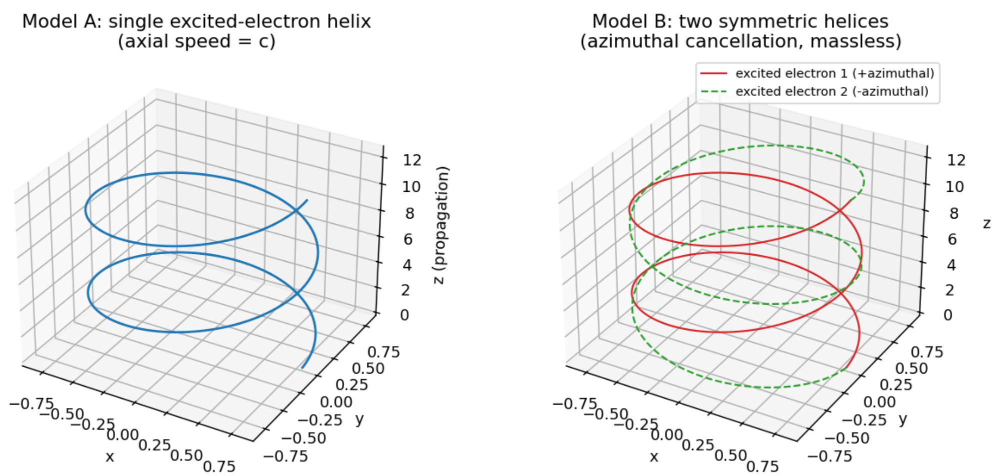

FIG. 4. Photon models. Left: model A, single excited-electron helix (axial speed = c). Right: model B, two symmetric helices — azimuthal cancellation leaves purely axial momentum.

VII. THE QUANTUM RELATIONS EMERGE FROM HELIX GEOMETRY

a. Energy as rotation (GQR). The Planck constant h is the angular momentum per turn; the reduced Planck constant $\hbar = h/2\pi$ is the per-radian angular momentum. The frequency is the number of turns per second (countable strands). Then

$$E = \hbar\omega = \frac{h}{2\pi} 2\pi f = h\, f = \underbrace{h}_{\text{angular momentum per turn}} \times \underbrace{f}_{\text{turns per second}}, \tag{4}$$

the Planck–Einstein relation [13] as the rotation energy of the helix, with the de Broglie wavelength [14] (Fig. 5).

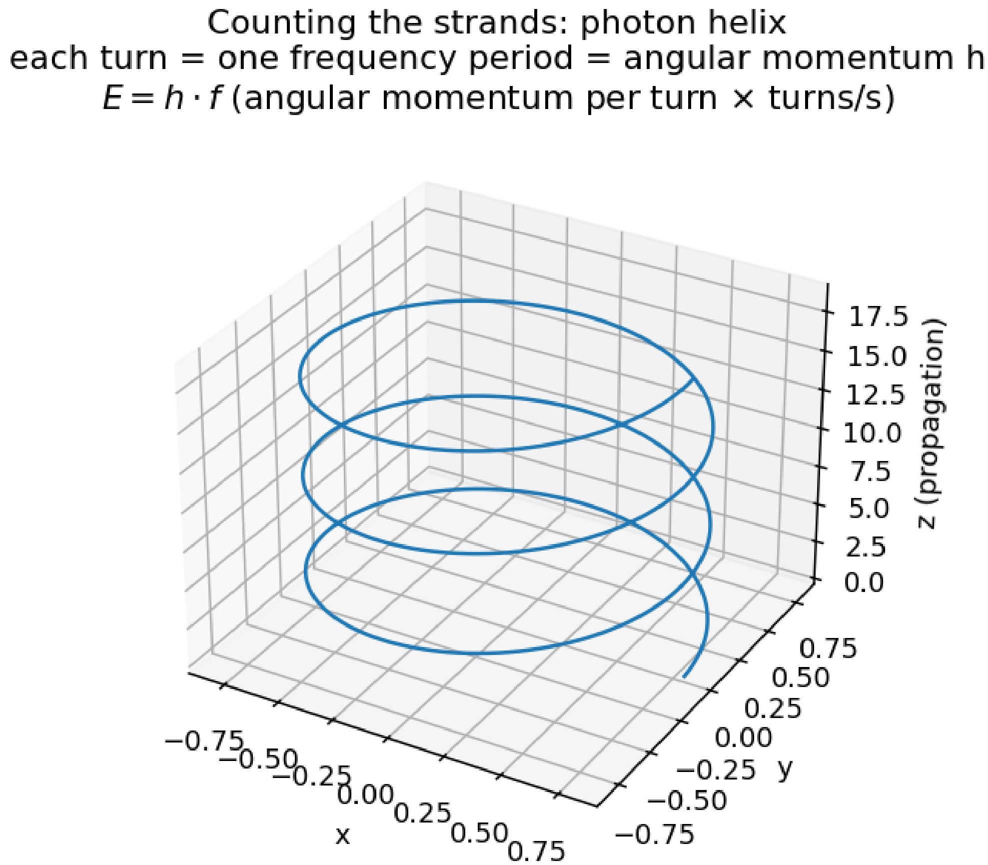

FIG. 5. Counting the strands: each turn of the photon helix is one frequency period and carries angular momentum h; $E = h \cdot f$ = angular momentum per turn × turns per second.

b. Circumference = wavelength (GQR4-6). The factor 2π between h and $\hbar$ is the circumference of the helix cross-section. With the geometric choice $r = \lambda/2\pi$ (the reduced wavelength as helix radius), the chain closes:

$$2\pi r = \lambda \qquad \text{(one turn = one wavelength),} \tag{5}$$

$$\hbar = r\,p \qquad \text{(angular momentum = radius × momentum),} \tag{6}$$

$$p = \frac{h}{\lambda} \qquad \text{(de Broglie relation emerges),} \tag{7}$$

$$E = pc = hf \qquad \text{(Planck–Einstein emerges),} \tag{8}$$

verified with physical constants at machine precision (Table I).

TABLE I. Circumference = wavelength: the derivation chain closed with physical constants (λ = 500 nm).

Quantity	Value	Check
$r = \lambda/2\pi$	79.58 nm	reduced wavelength
circumference $2\pi r$	500.0 nm	$= \lambda$ (one turn = one wavelength)
$p = h/\lambda$	1.3252×10^{-27} kg m s^{-1}	de Broglie
$\hbar = r\,p$	1.0546×10^{-34} J s	measured 1.05457×10^{-34}; rel. err. 6×10^{-10}
$E = pc$	3.313×10^{-19} J	$= hf$ exactly

c. Uncertainty from helix geometry (Heisenberg). The helix radius is the natural scale of positional uncertainty, $\Delta x \sim r = \lambda/2\pi$; the momentum is $\Delta p \sim p = h/\lambda$. Their product is

$$\Delta x\,\Delta p \;\sim\; \frac{\lambda}{2\pi}\,\frac{h}{\lambda} = \frac{h}{2\pi} = \hbar \;\geq\; \frac{\hbar}{2}, \tag{9}$$

so the Heisenberg uncertainty relation [15] emerges from the helix geometry at its natural scale (the helix cannot localize a photon better than its own radius without destroying the strand structure).

d. Exclusion of identical strands (Pauli). Two identical states are two parallel twistors; their symplectic volume vanishes, $\det(AA^\dagger) = |\det A|^2 = 0$ — identical strands create no new structure, no mass, no new degree of freedom. This is the geometric rank criterion for exclusion: only *distinct* strands (linearly independent twistors) open a new channel, in the spirit of the Pauli exclusion principle [16].

VIII. GLUEBALL, QUARK, AND THE ONE "3"

The glueball mass rule reads as strand counting: $m = \sqrt{N}\,M_0$ with $N = 3$ anchored directions ($\sigma_1\sigma_2\sigma_3$); the lattice N-sequence $\{3, 6, 7\}$ [17–19] (channels $0^{++}, 2^{++}, 0^{-+}$) matches $\sqrt{3}, \sqrt{6}, \sqrt{7}$ times M_0 within the lattice ranges. The number of colours is the same "3": colour = 3 = three spatial directions (structure correspondence: why 3 colours); 8 gluons $= 2^3 = \dim C\ell(6)$ (TW5). The "3" of space, of colour, and of glueball anchoring is one and the same structure.

Honest limit: the quark mass spectrum (2.2 MeV $\to$ 173 GeV) shows no strand-counting pattern — quark masses are QCD free parameters; the colour count gives structure, not numbers.

IX. NUMERICAL VALIDATION

All identities above are verified numerically (N1–N29; Lean 4 [5, 6], machine precision): the Cauchy–Binet chain ($\leq 5.9 \times 10^{-12}$), unitary transport (formula error 0, sub-volume conservation $\leq 5.9 \times 10^{-12}$), the light-cone bandwidth (violations 0), the four-force decomposition (error 0), the photon model (azimuthal cancellation 0), the quantum matching ($E = \hbar\omega$ relative error 0; $\hbar = rp$ relative error 6×10^{-10}; $E = pc$ vs $E = hf$ error 0), and the "3" chain ($\det_3 = 1$ for orthogonal unit directions, $m = \sqrt{3}\, M_0$).

X. HONEST ASSESSMENT

The following is claimed, and the following is not.

a. Claimed. (i) A single self-consistent chain from one primitive (flowing space) covers mass, information, causality, effective speed, force, the photon, and the quantum relations; (ii) the derivation chains are *closed*: given the geometric choice $r = \lambda/2\pi$, the de Broglie and Planck–Einstein relations are consequences of helix geometry plus classical identities; (iii) all statements are formalized in Lean 4 with zero `sorry` and verified numerically at machine precision.

b. Not claimed. (i) A first-principle derivation of the geometric choice $r = \lambda/2\pi$ (its origin is unknown); (ii) the numerical derivation of the remaining constants: G and e are *structurally present* — G is the flow-gradient constant in the rain speed $v(r) = \sqrt{2GM/r}$ (at the horizon $v = c$, exact), and e is the strand-flux quantum, with the fine-structure constant $\alpha = e^2/(4\pi\varepsilon_0\hbar c) = 1/137.036$ the geometric ratio of e, $\hbar$ and c (verified to 6×10^{-10}) — but their values still rest on the M_0 calibration and the measured α; (iii) any new falsifiable prediction beyond the standard framework (the lattice N-sequence is a posteriori fitting); (iv) the continuum wave equation for the photon's wave nature, and the divergence form of charge, remain formalization skeletons.

XI. CONCLUSION

The flowing-space framework unifies, in one chain of exact identities, the concepts that physics has kept separate: mass (symplectic volume), information (symplectic volume again), causality (lattice locality), force (Leibniz channels of flow momentum), the photon (excited-electron helix), and the quantum relations (rotation of the helix; circumference = wavelength). The chains are closed; the geometric choice at their head is the open question. Whether the foundation survives is a question for experiment; the framework is now precise enough to be falsified.

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